

SIF SENTINEL

Safety Assessment Report

Report ID:	SIH-DEMO-LIVE-01	Source File:	N/A
Location:	MCC Building 4 - Unit 12	Source Type:	synthetic
Date:	2026-09-23 21:08	Report Type:	near_miss

OBSERVATION

Contractor electrician accessed energized equipment without isolation during 4160V motor control center cubicle MCC-04 maintenance. LOTO not applied and isolation not verified prior to opening switchgear cabinet.

SAFETY ASSESSMENT

Activity	Maintenance
Hazard	Electrical Energy
Critical Barrier	Isolation
Barrier Status	Absent
Priority	PriorityLevel.HIGH

SAFETY SIGNIFICANCE

SIF Precursor Signal Probability: 47.7%

EVIDENCE

Energy Isolation Failure (CRITICAL)

Failure to isolate hazardous energy (electrical, mechanical, thermal, or hydraulic) before commencing work is a CRITICAL SIF precursor. Uncontrolled energy release can cause electrocution, crush injuries, or explosions. LOTO procedures must be verified and documented before any work on or near hazardous energy sources begins.

DECISION RATIONALE

Priority: HIGH Why: - Maintenance activity detected - Electrical Energy hazard detected - Isolation absent - Energy Isolation Failure safety rule triggered - Recurring risk pattern detected (similarity: 1.00) - Safety override applied: Elevated to HIGH: CRITICAL safety rule (severity 4/4) triggered. Key drivers: (1) Model identified moderate precursor indicators (probability: 47.7%); (2) Deterministic safety rules triggered: Energy Isolation Failure (severity: CRITICAL); (3) Safety barrier issue: Isolation is Absent involving Electrical Energy during Maintenance; (4) Elevated recurring risk pattern detected (similarity score: 1.00); (5) Safety policy override: Elevated to HIGH: CRITICAL safety rule (severity 4/4) triggered.. Trigger evidence: 'loto not applied, isolation not verified, energized equipment, without isolation'. Note: This score prioritizes HSE review attention and does not predict accident occurrence.

ACTION STATUS

Current Workflow State: HSE_REVIEW_REQUIRED

AUDIT TRAIL

- 2026-09-23T21:08 - ANALYZE_REPORT by system
- 2026-09-23T21:08 - ALERT_TRIGGERED by SIF_SENTINEL_AUTOMATION

AI-assisted safety triage. This assessment identifies potential precursor signals and does not predict accidents. Final safety decisions remain subject to HSE review.

SYNTHETIC / DEMONSTRATION DATA